

I Proposed a Desalination Plant Concept Before It Was Built

What happened 13 years later?

A retrospective on engineering, infrastructure, and the limits of technical solutions

2013 thesis → 2015 plant → 2025 shutdown

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Analysis of Power and Water Cogeneration:
A Case Study in Accra, Ghana



Thesis, Environmental Engineering & Management

Prof. Dr. Kraft, Prof. Dr. Jentsch,
Bauhaus University Weimar

The retrospective question

Not “was I right?” — but what does reality reveal about the model?

Analysis of Power and Water Cogeneration:
in Accra, Ghana



sis, Environmental Engineering & Management

Craft, Prof. Dr. Jentsch,
auhaus University Weimar

2013

A master's thesis examined a proposed 60,000 m³/day desalination plant near Accra and asked whether water and energy infrastructure could be planned together.

2015–2025

The plant was built. It supplied water. It also became financially and institutionally troubled.

This video asks what changed when an engineering concept became an infrastructure history.

The original problem was bigger than a plant

Accra's water supply issue was tied to demand growth, energy, and governance.

ground:

Water Company Limited is planning new supply and treatment facilities near Accra, Ghana. This includes a 100 m³/day desalination facility which will be the first large-scale desalination contract ever to be signed in West Africa, a number of unique technical and institutional questions:



1. Is **desalination needed** in a country such as Ghana, with access to freshwater sources?

2. Which **desalination method** best suits circumstances and Best Available Technology (BAT)?

3. Is there **sufficient energy** for the water supply facility; which type of power facility would be most appropriate?

4. Is **desalination the best alternative** to meet human needs for water?

5. Does the **Environmental Impact Assessment (EIA)** law requires project justification and assessment of the possible alternatives to the initiative.

Water demand was rising

GAMA growth and supply shortfalls

Energy demand mattered

treatment, pumping, and transport

Policy context mattered

EIA, alternatives, affordability

The thesis began with a technical proposal, but the actual question was: which problem are we solving?

Desalination looked technically plausible

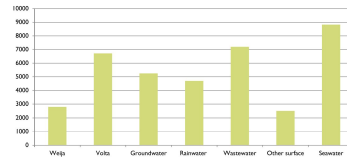
The old model gave seawater a strong score because it diversified supply.

Selection of Optimal Water Supply

	Capacity	Comments
	273,000m ³ /day	Current withdrawals of 170,000m ³ /day are approaching capacity; Pollution is concern.
	90,400,000m ³ /day	Plenty of supply; Transport over 50km, leads to higher costs; overreliance on one source raises water security concerns.
Water	100,000m ³ /day	Brackish groundwater would require desalination. Large groundwater withdrawals generally not possible.
Water Harvesting	55,000 m ³ /day	Not suitable on large scale; treatment issues if used as drinking water.
Water	500,000 m ³ /day	Current WWT and storm water infrastructure is inadequate. Waste water withdrawal from coastal lagoons is brackish.

Source: adapted from SWITCH EU, 2011

Weighted Product Model



Seawater is judged the optimal supply solution due to its abundant volume, low risk of contamination, and the fact that it diversifies existing sources providing security of supply.

Lake Volta is likely the optimal supply *ceteris paribus*, but in this particular time and context, considering water security concerns and heavy reliance on Kpong sources, desalination was given a preferential rating.

Why seawater scored well

- Abundant source**
not dependent on river yield
- Low contamination risk**
compared with some local sources
- Supply diversification**
security against one-source dependence

But the model still had to ask: security for whom, and under what institutional conditions?

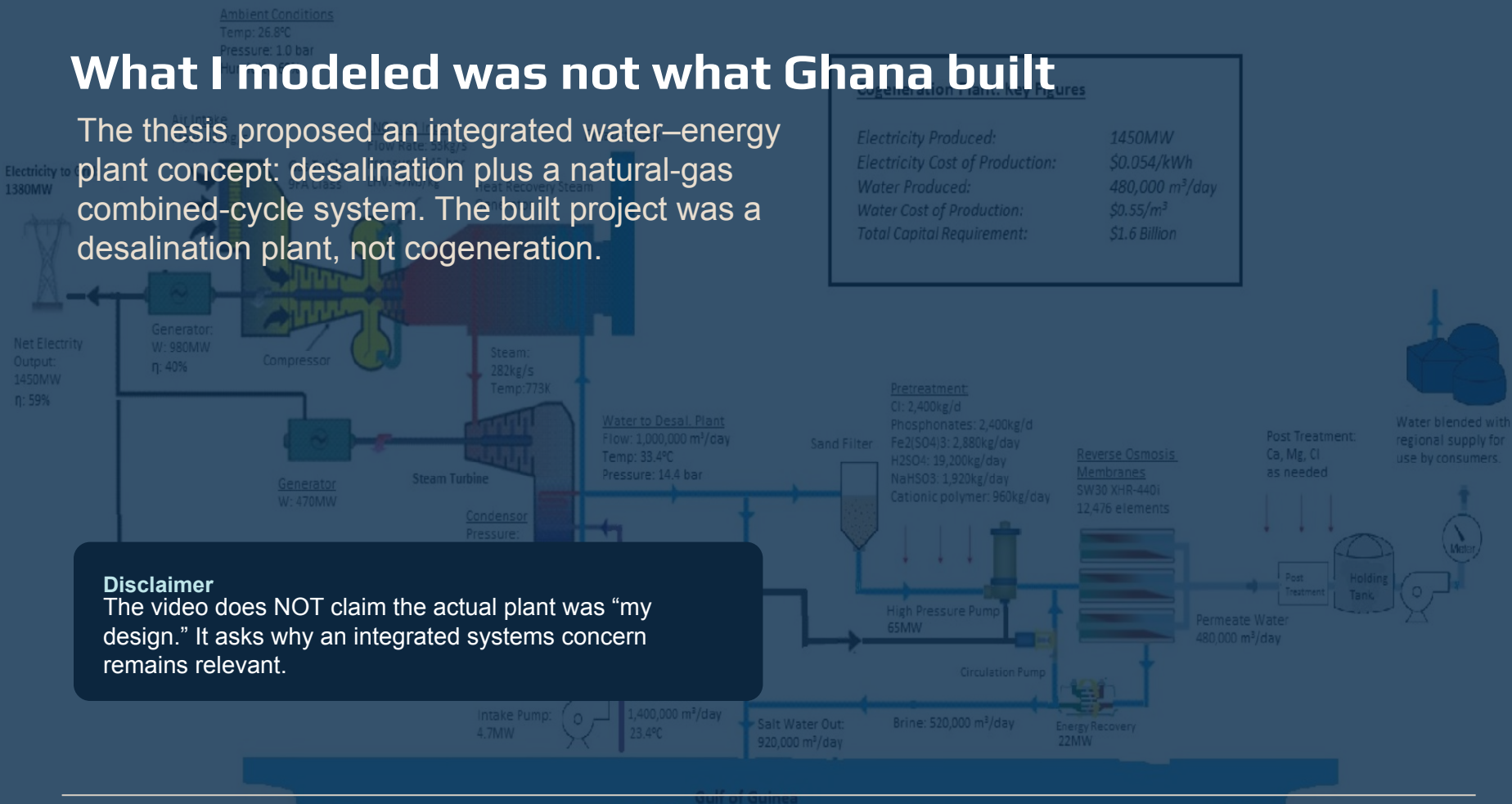
What I modeled was not what Ghana built

The thesis proposed an integrated water–energy plant concept: desalination plus a natural-gas combined-cycle system. The built project was a desalination plant, not cogeneration.

Electricity Produced:	1450MW
Electricity Cost of Production:	\$0.054/kWh
Water Produced:	480,000 m ³ /day
Water Cost of Production:	\$0.55/m ³
Total Capital Requirement:	\$1.6 Billion

Disclaimer

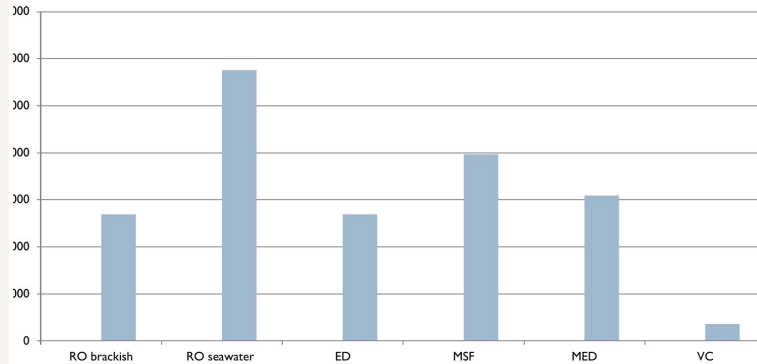
The video does NOT claim the actual plant was “my design.” It asks why an integrated systems concern remains relevant.



The technology choice aged reasonably well

Reverse osmosis remained the practical desalination pathway.

Desalination: Multi-Criteria Analysis



Reverse Osmosis of seawater is the optimal technology in this case. This conclusion is consistent with similar multi-criteria analyses applied to cogeneration of water and power. For example, a study to determine the optimal desalination method for an energy-rich Arabian Gulf region concluded that reverse osmosis was most appropriate, despite MSF systems being most commonly built in the region.

Source: Hajeer & Othman, 2005

Plant Concept

Power plant operating parameters	
Cooling media	Seawater
Water quantity (60-100x exhaust steam flow)	1,413,542.37 m ³ /day (58 x steam flow)
Fuel LHV	47 MJ/kg
Gas supply rate	53 kg/s
Efficiency	59%
Net output	1,426.11 MW

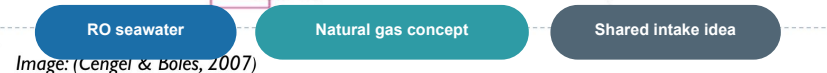
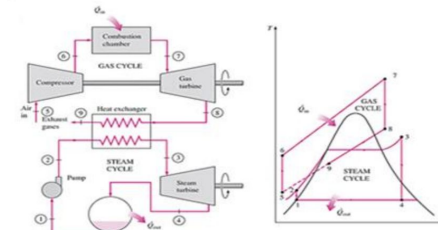


Image: (Cengel & Boles, 2007)

What held up: desalination was technically feasible. What remained uncertain: the economics and institutional design around it.

A production-cost model is not a contract

The hard part turned out not to be only membrane efficiency.

Production Cost of Water

Capital Cost Estimate	Cost \$USD
Preparation Cprep	3,000,000
Pre	0 (included in gas plant cost)
Treatment	24,600,000
System Equipment	111,000,000
Treatment	6,200,000
Concentrate Disposal	0 (included in gas plant cost)
Wastewater Management	4,600,000
Electrical System	7,700,000
Product Storage & Distribution	6,000,000
Buildings	15,400,000
Setup Costs	6,200,000
Project engineering services	33,900,000
Administration, contracting, management	7,700,000
Environmental permitting	0 (included in gas plant cost)
Utilities services	0 (included in gas plant cost)
Project financing cost	37,000,000
Agency	20,000,000
TOTAL CAPITAL COSTS	283,000,000
Cost \$/m3	\$590.00

Operation Cost Estimate	Cost \$USD/y
Operation & Maintenance	1,826,400
Replacement Parts & Materials	14,268,000
Replacement Membranes	6,086,400
Power	30,747,600
Chemicals	14,500,800
TOTAL	\$28,535,160

The total unit cost of water is the sum of operation and maintenance costs plus the amortized capital (over 25 years at 5% interest).

Unit Cost = \$0.55/m³

Modeled production cost

\$0.55/m³

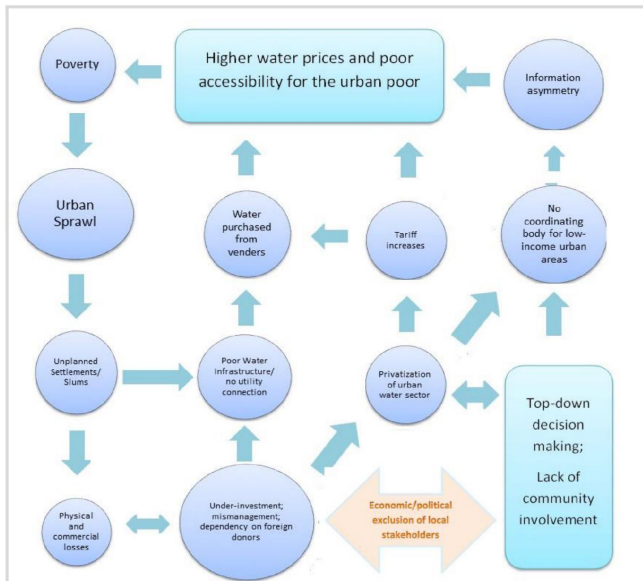
Real-world infrastructure added

- financing and guarantees
- bulk-water purchase agreement
- maintenance obligations
- tariffs, payment delays, political risk

The most important slide in the old presentation

More water production does not automatically solve affordability or access.

Water Affordability and Access



Problem diagram for lack of accessibility and affordability of potable water in the urban area of Greater Accra.

Top down decision making and lack of community involvement as described by (Fuest & Haffner, 2010) is seen as an ongoing management pattern that framed human decisions leading to the current problems.

A desalination plant does NOT solve this problem.

A desalination plant does not solve:

● **network access**

who is actually connected?

● **tariff burden**

who pays and how much?

● **utility losses**

leaks, illegal connections, outages

● **governance**

coordination, maintenance, contracts

This is where the project stopped being only an engineering problem.

What happened after the thesis?

A decade-long history makes the retrospective possible.



The plant was not simply a failure. It supplied water. But each shutdown revealed how technical capacity depends on contracts, maintenance, and payment systems.

So, did it work?

Only if “work” is split into different kinds of success.



Technical

Produced potable water

yes, when operating



Service delivery

Improved water availability

real community benefit



Financial

Sustainable public cost

contested / strained



Institutional

Stable contract + maintenance

recurring challenge



Distributive

Solves access for poorest?

not by production alone

The retrospective is stronger if it avoids a simple verdict.

The hidden variable: institutional reliability

Security of supply is not just physical diversification.

2013 model

**More sources =
more resilience**

seawater diversifies away from Volta/Weija
dependence



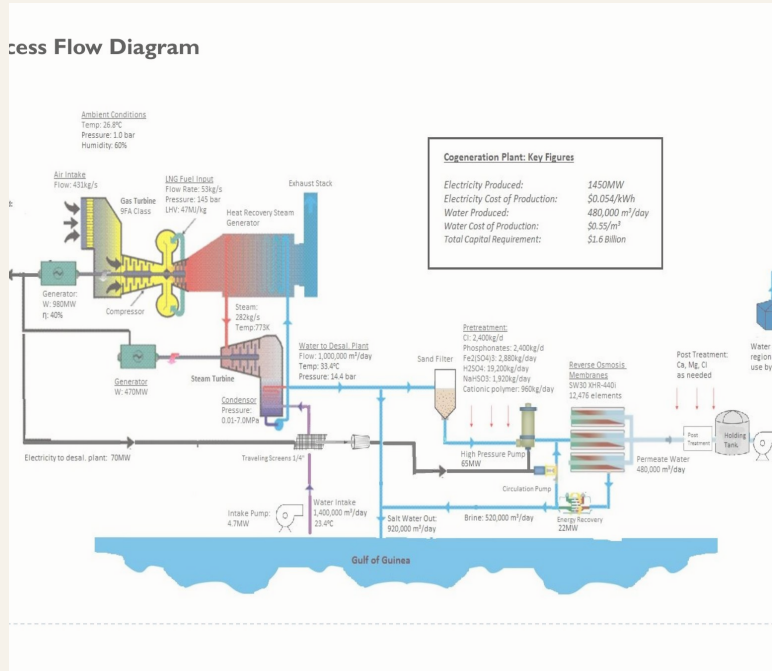
2026 reflection

**A new source can
create new
dependencies**

energy, contracts, payment flows, maintenance,
spare parts

**A desalination plant can be physically independent of a river and still
institutionally fragile.**

Why this pushed me beyond conventional engineering



The engineering was real.

Mass flows. Energy balances. Reverse osmosis. Cost estimation. Process diagrams.

But the hardest questions began where the calculation ended.

Who pays? Who is connected? Who maintains it? Who benefits? What counts as “optimal”?

The question was never only:

Can we desalinate seawater?

We can.

The harder question is:

What kind of system has to surround the technology for it to solve the problem we actually care about?

13 years later, the technology still matters, but the system matters more.

Key Conclusions

Desalination of seawater on a large scale is possible in the Greater Accra region and seawater is an abundant, affordable supply source for most (though not all) consumers.

The cost of desalination remains significantly higher than the main freshwater source, Lake Volta. Lake Volta will likely continue to be the main potable water source for the growing metropolis. Desalination would add security of supply.

If a desalination plant is built, cogeneration with a natural gas plant could present cost savings and improved system performance.

System upgrades, such as reducing water system losses and electrical transmission losses, should take priority over any supply expansions (though both will be needed within the next 10 years).

Additional institutional reforms are needed to allow the urban poor to benefit from any additional water supply infrastructure (outlined in Ch. 5);

Desalination is not the answer to the underlying issues in the urban water sector, which are primarily institutional and social, rather than technical problems.